Audio Encryption and Decryption

Using the 5D tangent-hyperbolic memristive chaotic model

Main outcome: Correct-key decryption recovered every original PCM sample exactly. MSE = 0 and SNR = ∞ dB.

1. Objective

The aim is to demonstrate audio encryption and decryption using a pseudorandom keystream generated by the five-dimensional chaotic model. The audio samples are XORed with the chaotic keystream. Applying the same keystream again reverses the XOR operation and restores the original samples.

2. Test audio data

Property	Actual input
Input file	test_sample-6s.wav
Duration	6.3913 seconds
Sampling rate	44,100 Hz
Channels	2 (stereo)
Format	16-bit signed PCM WAV
Samples per channel	281,856
Sample range	-22298 to 18381

3. Chaotic model and parameters

The model has five state variables. The first equation below uses the parameter placement from the article. The notebook uses classical fourth-order Runge–Kutta (RK4), so this audio experiment is the integer-order case $\alpha = 1$.

$$\dot{x}_1 = a(x_2 - x_1) + 0.25x_3 + g(\delta + \gamma \tanh(x_5))x_3$$
$$\dot{x}_2 = bx_1 - x_1x_3 + cx_2 - x_4$$
$$\dot{x}_3 = dx_1x_2 - ex_3$$
$$\dot{x}_4 = x_1$$
$$\dot{x}_5 = x_3$$

Setting	Value
a, b, c, d, e	78, −16, 58, 35, 15
g, δ, γ	1, 1, 0.5
Initial state [x_1, x_2, x_3, x_4, x_5]	[0, 0, 8, 6, 0]
RK4 time step h	0.001
Transient discarded	10,000 RK4 steps
Steps between audio keystream samples	10

4. Encryption and decryption flow

1. Read the WAV: Load the audio and preserve its sampling rate, stereo channels and int16 PCM sample format.

2. Initialize the 5D system: Set the paper parameters and initial conditions. Integrate the model with RK4 and discard the initial transient.

3. Generate chaotic values: For each audio sample index, advance the system by 10 RK4 steps and take x_1 and x_2 for the two stereo channels.

4. Convert to a 16-bit keystream: For each channel value x, compute K = floor($|x| \times 10^8$) modulo 65,536, then store it as uint16.

5. Encrypt: View each signed int16 sample as a uint16 word and XOR it with the corresponding chaotic keystream word.

6. Decrypt: Restart the same chaotic system from the same initial conditions, regenerate the same keystream and XOR it with the encrypted samples.

7. Verify and save: Compare recovered samples with the original samples, calculate error/statistics and save the encrypted and decrypted WAV files.

5. Core encryption equations

$$\mathbf{K[n, c] = floor(|x^c[n]| \times 10^8) \bmod 65{,}536}$$

$$\mathbf{C[n, c] = P[n, c] \text{ XOR } K[n, c]}$$

$$\mathbf{\hat{P}[n, c] = C[n, c] \text{ XOR } K[n, c]}$$

P is the original sample, K is the chaotic 16-bit keystream, C is the encrypted sample and $\hat{P}$ is the recovered sample. XOR is reversible because (P XOR K) XOR K = P.

6. Actual output: waveform comparison

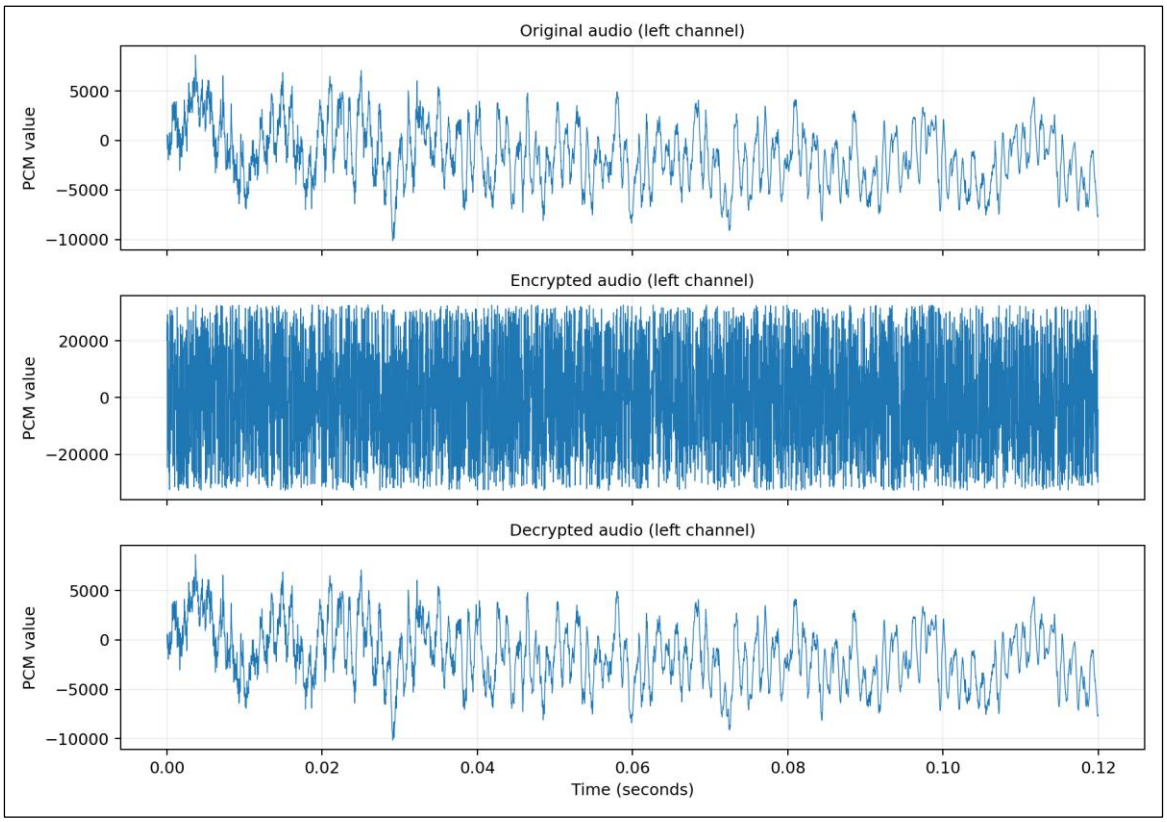


Figure 1. First 0.12 seconds of the left channel: original, encrypted and decrypted waveforms. The decrypted waveform overlaps the original.

7. Actual output: byte histograms

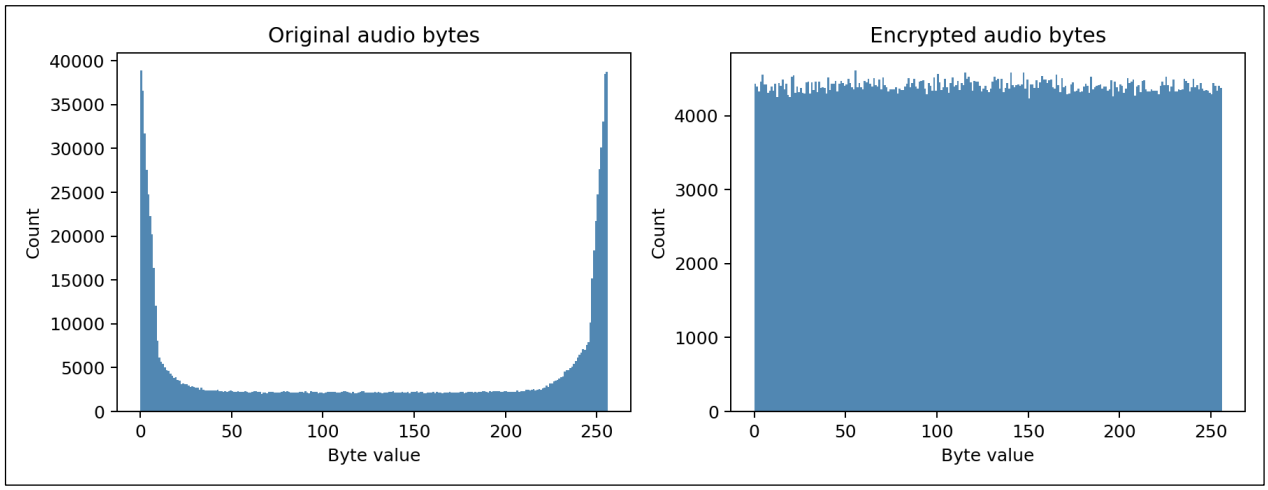


Figure 2. Histogram of the original and encrypted audio bytes. The encrypted byte distribution is much flatter in this sample.

8. Test results

Test	Measured result	Interpretation
Exact sample recovery	True	PASS — every sample equals the original
MSE (correct key)	0.0	PASS — zero reconstruction error
SNR (correct key)	∞ dB	PASS — no sample error
Original byte entropy	7.2045 bits	Measured on all bytes of the int16 WAV data
Encrypted byte entropy	7.9998 bits	Close to the 8-bit maximum; statistical observation only
Original lag-1 correlation (L/R)	0.9902 / 0.9902	High adjacent-sample correlation in this recording
Encrypted lag-1 correlation (L/R)	-0.0025 / -0.0020	Near zero in this test
Wrong-key MSE	362900295.52	Wrong-key output differs from the original
Wrong-key SNR	-16.75 dB	Decryption fails with changed initial condition

9. Wrong-key test

For the wrong-key demonstration, the first initial condition was changed from $x_1(0) = 0$ to $x_1(0) = 10^{-10}$ while the other settings remained unchanged. The wrong-key decryption did not recover the original audio: MSE = 362,900,295.52 and SNR = −16.75 dB.

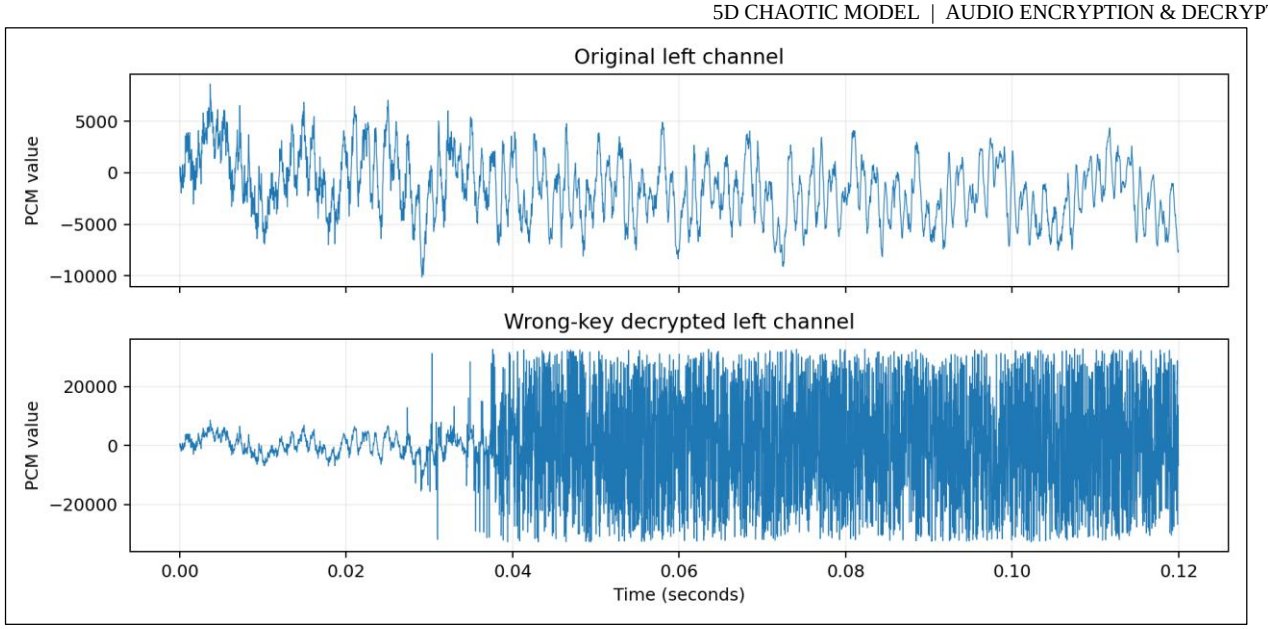


Figure 3. Original and wrong-key decrypted left-channel waveform over the first 0.12 seconds.

10. Spectrogram comparison

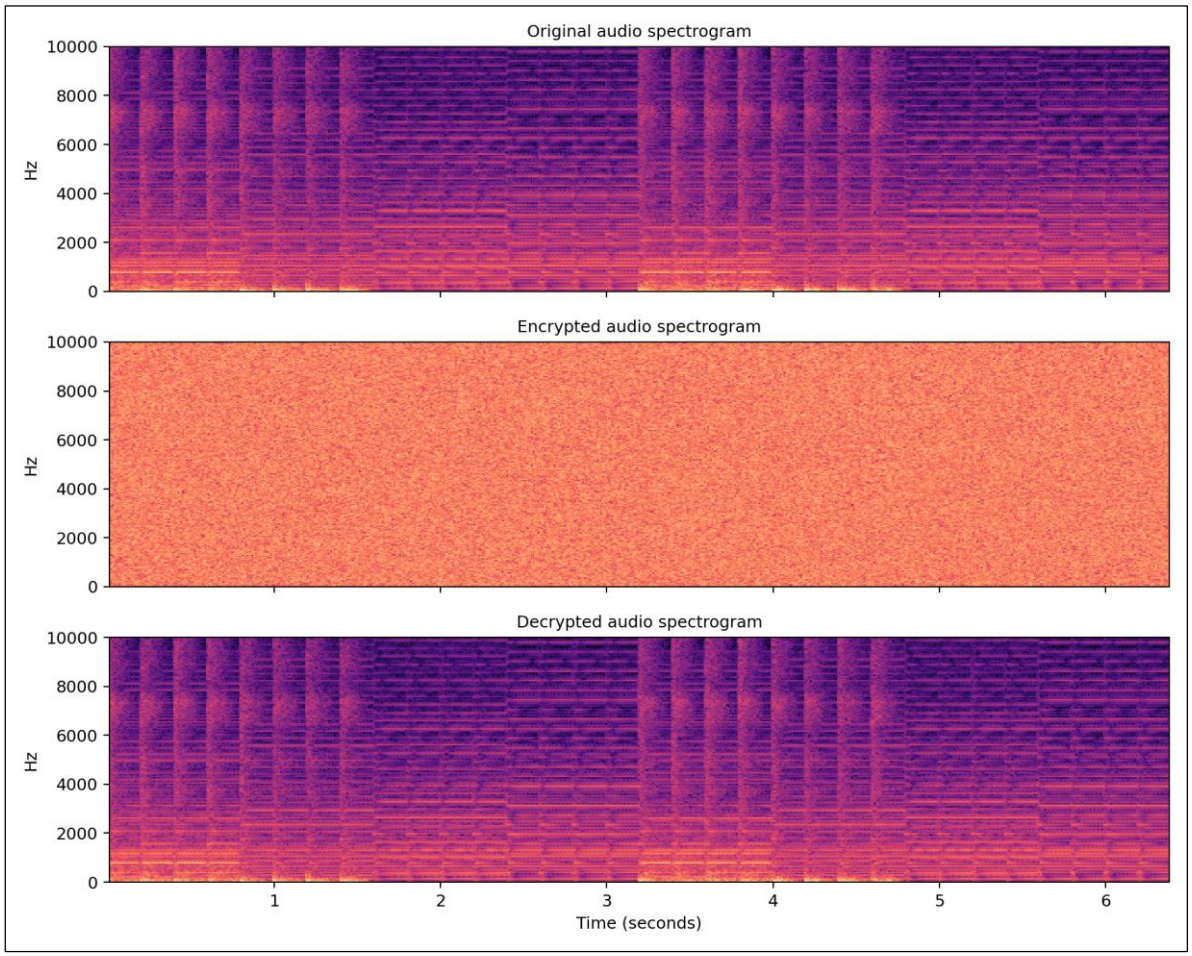


Figure 4. Original, encrypted and decrypted audio spectrograms. The encrypted spectrogram appears noise-like; the recovered spectrogram matches the original.

11. Code flow in simple words

- The program reads the WAV file and checks that it is stereo, 16-bit PCM.
- It creates the 5D chaotic system with the fixed model parameters and initial conditions.
- After removing the transient, it produces two chaotic values per sample index: x_1 for the left channel and x_2 for the right channel.
- The values are converted into 16-bit integers and used as the XOR mask.
- The encrypted samples are written to a new WAV file. Decryption regenerates the same mask and applies XOR again.
- Finally, the program checks sample-by-sample equality and computes entropy, adjacent-sample correlation, MSE and SNR.

12. Important limitations and interpretation

- The 2024 article presents its AIE application for images. This exact audio XOR pipeline is a project extension; it should not be described as the audio algorithm demonstrated in that article.
- This notebook uses RK4 with $\alpha = 1$. It does not implement the paper's ABC fractional-order numerical scheme for $\alpha = 0.99$.
- No separate key-audio file was supplied for this experiment. Therefore, only the chaotic keystream was used; the optional second audio key was not used.
- The method is XOR-only and has no diffusion stage. High entropy and low adjacent-sample correlation do not prove cryptographic security. A single changed plaintext sample would affect only the corresponding ciphertext sample.
- The chaotic stream is deterministic and restarts from the same key. Reusing the same keystream for different recordings can reveal relationships between the plaintexts. Do not claim production-grade security.

13. Conclusion

The 5D chaotic model generated a keystream that transformed the uploaded audio into an encrypted-looking PCM signal. Using the same parameters and initial conditions regenerated the same keystream and recovered all 281,856 stereo sample frames exactly. The zero MSE confirms a lossless round trip for this experiment. The wrong-key test failed to recover the original, as expected. These results demonstrate functional reversibility and statistical changes in the tested signal, but do not establish cryptographic security.

Reference

Qureshi, M. A., Khan, N. A., Raza, S., & Iqbal, S. M. Z. (2024). "Cap like trajectories in 5D chaotic tangent hyperbolic memristive model: fractional calculus and encryption." Physica Scripta, 99, 075238. https://doi.org/10.1088/1402-4896/ad5476

Github link:

https://github.com/fizza-14/5D-Chaotic-Memristive-Image-Audio-Video-Encryption-Decryption